DSA0308-NATURAL LANGUAGE PROCESSING

ASSIGNMENT 4

CO4: SEMANTIC ANALYSIS, WSD AND RETRIEVAL

COURSE: NATURAL LANGUAGE PROCESSING

NATURE: INDIVIDUAL ASSIGNMENT

TOTAL MARKS: 50

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REGISTER NUMBER: 192424240

CONTENTS AND SUBMISSION NOTE

This report answers the tasks in the corresponding uploaded assignment. It follows the task order and preserves the assignment's terminology. The worked examples explain the reasoning and the evidence expected.

Important: where the assignment requires a corpus experiment, annotation, code execution or human judgement, no unrun result is presented as an actual measurement. Such values must be collected by running the experiment and inserted into the report. Any sample calculation is explanatory only.

Part I Task 1 Meaning Representation

Concepts and direct answer

For 'Maya gave Ravi a book yesterday', identify giver/agent Maya, recipient Ravi, theme book, giving event and time yesterday. Represent as GIVE(e1), AGENT(e1,Maya), RECIPIENT(e1,Ravi), THEME(e1,Book), TIME(e1,yesterday). Also create a frame with event slots. Explain how passive paraphrase preserves roles despite different word order.

The answer begins by identifying the input, the target interpretation and the assumptions. Terms are defined before they are used. This prevents the same label from being applied to different meanings in different parts of the analysis.

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Part I Task 1 Meaning Representation

Representation and method

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Evaluation, limitations and conclusion

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Part I Task 2 First-Order Predicate Calculus

Concepts and direct answer

Define Student(x), Course(c), Instructor(i), Enrolled(x,c), Teaches(i,c) and Passed(x,c). Formalise every student enrolled in at least one course; some student passed every course taught by Dr Rao; no instructor teaches every course. Explain quantifier scope and show a finite countermodel where forall-exists differs from exists-forall. Check consistency of universal and existential-negative statements.

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Part I Task 3 Syntax-Driven Semantics

Concepts and direct answer

For 'The experienced researcher saw the student with the telescope', identify entities, event, modifier and possible instrument/accompaniment. Compare PP attachment to VP (researcher used telescope) and NP (student had telescope). Trace composition from word meanings to sentence representation. Validate role compatibility and passive-role assignment.

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Part I Task 4 Lexemes and WSD

Concepts and direct answer

Distinguish lexeme, word form and sense. 'Bank' in deposited cash is financial noun; river bank is geographical noun; banked the aircraft is an aviation verb. Design WSD using context window, POS, collocations and candidate senses. Apply to three supplied examples and one new sentence. Evaluate accuracy and discuss imbalance and insufficient context.

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Part I Task 5 Semantic Retrieval

Concepts and direct answer

Represent query 'methods for finding the meaning of ambiguous words' as word-sense-disambiguation intent. Compare keyword matching with semantic retrieval. Rank using lexical relevance, concept similarity and sense match. Define precision and recall, calculate on a small collection and discuss false positives and false negatives.

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Part I Task 6 Integrated Challenge

Concepts and direct answer

For 'I saw her duck near the bank, and it was closed', analyse duck noun/verb, bank financial/river, her role and it antecedent. Construct at least two semantic structures using event frames or FOPC. Use syntax and context to select a reading, build a retrieval query and rank articles. Retain alternatives or ask clarification when evidence is insufficient.

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Part II Task 1 Event and Role Representation

Concepts and direct answer

For 'Leena sent the report to Arun after the meeting', represent SEND event, agent, theme, recipient and AFTER relation. Provide semantic-role frame and graph. Give a meaning-preserving passive rewrite and a validation rule for missing participants.

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Part II Task 2 Formalise and Test

Concepts and direct answer

Define Learner, Course, Enrolled, Completed, Prerequisite and OfferedBy. Formalise all three statements from the assignment. Explain scope, build a finite model, and construct a small knowledge base with facts and a rule. Distinguish failure to prove from proof of negation under an open-world assumption.

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Part II Task 3 Parse-to-Meaning

Concepts and direct answer

For 'The analyst noticed the student with the camera near the library', show two syntactic attachments and semantic composition. Add type/role checks and an adversarial example where a syntactically valid parse is semantically implausible. Refine with context and selectional preferences.

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Part II Task 4 Light WSD

Concepts and direct answer

Create sense inventory for light: illumination noun, low-weight adjective and light meal/low-calorie adjective. Specify input, candidate senses, features and decision rule. Apply to supplied and original examples; include a likely failure. Create at least eight labelled instances and calculate accuracy with per-sense interpretation.

The answer begins by identifying the input, the target interpretation and the assumptions. Terms are defined before they are used. This prevents the same label from being applied to different meanings in different parts of the analysis.

The main conclusion is that the system must make its intermediate decisions explicit. A final label or output alone is not enough to show that the method is correct.

Evidence	Required content
Input	Sentence, corpus, query or scenario
Method	Formula, grammar, rule or algorithm
Trace	Intermediate steps and selected output
Validation	Metric, counterexample or comparison

Part II Task 4 Light WSD

Representation and method

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A reliable design separates input preparation, representation, decision procedure and output. Each stage should have a stated data structure and a clear rule for updating it. Keep the original input so that normalisation does not erase information needed later.

For reproducibility, record the data source, assumptions, parameter values, software version and fixed random seed wherever applicable.

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Part II Task 4 Light WSD

Worked example / trace

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Worked trace template: (1) identify the input units; (2) list candidate interpretations or structures; (3) apply the declared rule; (4) record the selected result and any unresolved alternative; (5) verify the result against a hand-worked example or gold annotation.

When two interpretations remain plausible, retain both or ask a clarification question. A forced choice without evidence can create a downstream error.

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Part II Task 4 Light WSD

Evaluation, limitations and conclusion

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Evaluation should include a normal case, a boundary case and an adversarial case. Report expected behaviour, actual behaviour, error category and a specific refinement. Use the same test inputs when comparing two designs.

Conclusion: select a design only when its measured behaviour meets the task requirements. Discuss remaining limitations and the conditions under which a simpler alternative may be preferable.

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Part II Task 5 Semantic Search

Concepts and direct answer

Represent query 'How can a system identify the intended meaning of a word in context?' Create at least five candidate documents, including paraphrases and distractors. Define a weighted score using semantic similarity, concept coverage, lexical overlap and sense compatibility. Calculate rankings, precision and recall at a stated cutoff.

The answer begins by identifying the input, the target interpretation and the assumptions. Terms are defined before they are used. This prevents the same label from being applied to different meanings in different parts of the analysis.

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Part II Task 6 Crane and Bank Challenge

Concepts and direct answer

For 'I watched the crane by the bank, then it moved', compare crane bird/machine and bank river/financial senses. Create candidate meaning graphs, resolve it using salience and semantic compatibility, and rank at least four documents. Include a lexical distractor and a refinement for insufficient evidence.

The answer begins by identifying the input, the target interpretation and the assumptions. Terms are defined before they are used. This prevents the same label from being applied to different meanings in different parts of the analysis.

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Additional Analysis – Engineering trade-offs

Discuss accuracy, memory, latency, transparency, annotation cost and maintainability. A method is useful only relative to its operating constraints.

Application to this assignment: CO4: SEMANTIC ANALYSIS, WSD AND RETRIEVAL. The design should preserve the input, expose intermediate decisions, state uncertainty and allow each stage to be checked independently.

Suggested evidence record: Input | Expected result | Actual result | Error category | Proposed correction. Complete this record using the student's own execution or annotation; do not present an illustrative value as a measured result.

CONCLUSION

The answers in this report follow the structure and concepts of the uploaded assignment. The central engineering lesson is to define representations and decision rules, trace the reasoning, test difficult cases, measure performance and defend the final design with evidence.

REFERENCES AND ACKNOWLEDGEMENT

Primary source: the corresponding DSA03 Natural Language Processing assignment PDF supplied for this task. Add exact textbooks, papers, corpora, toolkits and software references actually consulted, including author or organisation, year, title and publication details.

Disclose AI or reference tools used as required by the assignment. Verify the examples and replace all illustrative experimental values with results from your own implementation, corpus analysis or annotation.